

Fig. 5.6 Evolution of the line segment of initial conditions, for $t = 4, 8, 20, 28, 36$; the first four images are recorded in the *leftmost panel* and successive images are ordered *left to right*

becoming extended, and folding back on itself in an increasingly complex manner. Eventually the points fill in a region in the plane which is to all appearances identical to that obtained as the limit of a single long trajectory shown in the right panel of Fig. 1.24. (In these figures we chose $\varepsilon = 0.5$ and $\mathrm{E}_0 = 0.21048$, which matches the energy observed in our earlier simulations.)

The segment of initial conditions, a one-dimensional object, when transformed by the smooth map defined by solving initial value problems, should result, again, in a one-dimensional object, i.e. a curve. Yet it appears also to provide a dense covering of a two-dimensional region in the position projection. Given the presence of the momenta, the energy surface $H = \mathrm{E}_0$ is generically three-dimensional, since it is the level set of a function defined on $\mathbb{R}^4$. A more detailed numerical investigation would reveal that the trajectories of the differential equations themselves appear to generate, in the limit, three-dimensional space-filling curves.

Now consider an alternative initial set. Begin, as before, with a line segment on the x-axis, but now choose the initial momenta from the uniform distribution on the circle $p_x^2 + p_y^2 = 2(\mathrm{E}_0 - U(x, 0))$. In this way we again get initial conditions on the $H = \mathrm{E}_0$ energy surface, but this time the points are not close to one another in phase space, since the momenta of successive points may differ greatly. Integrate the set forward in time by solving the differential equations, and examine the image sets at successive instants in time. The leftmost panel in Fig. 5.7 shows the evolution of the initial interval at times $t = 0, 0.2, 0.4, \ldots, 1.8$. In the projection onto positions, the set now spreads out in a series of uneven concentric bands. The structure of these bands becomes increasingly complex and filamented and eventually spreads around the accessible region at the given energy. Snapshots at $t = 8, 16, 24$ are shown in Fig. 5.7.

A small set of initial conditions is eventually mapped forward by solving the differential equation into what appears to be a dense covering of the same region seen already in the right panel of Fig. 1.24 (which arose from the very long term evolution of a single initial condition).

As we see in the above example, the early evolution of the set is strongly dependent on the initialization, but in the long term, a similar region is ultimately covered (at least to visual inspection). However, also note that the approach to equilibrium is very different for the two choices of initialization, as we can see from